St. Philomena's College (Autonomous), Mysore

PG Department of Studies and Research in Chemistry

Question Bank {Revised Curriculum (LOCF) - 2020-22 Batch }

SecondYear –Third Semester

Course Title (Paper Title): CHEMISTRY OF FOOD AND BEVERAGES

QP Code: 85343

Unit	SI. No	Questions	Marks
1	1	What is food chemistry? Mention any two major and minor constituent of food.	2
1	2	Name three scientists who have done pioneering work in the development of food chemistry.	2
1	3	What is a balanced diet?	2
1	4	Write any two differences between energy yielding and body building food.	2
1	5	How protective food has been classified? Write its example	2
1	6	Write any four physical properties of water.	2
1	7	Write the importance and function of water in human body.	2
1	8	Write the Graph of sorption isothermal for typical food products showing the hysteresis.	2
1	9	Why quick freezing better than slow freezing?	2
1	10	What is meant by water activity? How its differ with moisture..	2
1	11	Name the three forms of water present in food.	2
1	12	Write any four function of water in food.	2
1	13	Mention any two role of A_w in food processing.	2
1	14	What is meant by water activity? Mention any two methods used for removal water from food.	2
1	15	What are the advantages of vaccume drying over conventional oven drying method?	2
1	16	Explain the changes arise due to effect of heat on the nutrient like carbohydrates present in food?	2
1	17	Explain the changes arise due to effect of heat on the nutrient like fat present in food?	2
1	18	Explain the changes arise due to effect of heat on the nutrient like protein present in food?	2
1	19	Explain the changes arise due to effect of heat on the nutrient like vitamins present in food?	2
1	20	Explain the changes arise due to effect of heat on the nutrient like vitamins and minerals present in food?	2
1	21	What are colloids? Write it types	2
1	22	Given any two examples of digestible and indigestible forms of carbohydrates	2
1	23	Write any four causes of food when food is get cooked.	2

1	24	Write the structure of aldose and ketose form of Glucose.	2
1	25	Define enantiomer with an example.	2
1	26	Write the boat and chair conformation α-D-Glucose.	2
1	27	What is the difference between cellulose and starch?	2
1	28	What is meant by cramilazation? Explain	2
1	29	What are carbohydrates? How its classified?	2
1	30	What are application of osmosis in food processing techniques.?	2
1	31	Write the dispersed phase and continuous phase form in sol and emulsion with an example each.	2
1	32	Write the dispersed phase and continuous phase form in gel and foam with an example each.	2
1	33	Write the Fischer and Haworth form of Glucose.	2
1	34	What is the difference between cellulose and starch?	2
1	35	Write the open chain structure of glucose, Ribose, lyxose and fructose.	2
1	36	Name the different groups of amino acids.	2
1	37	How proteins are important to our body.	2
1	38	Define albumins and globulins.	2
1	39	What are phosphoproteins and lipoproteins?	2
1	40	What are the major difference between proteins of animals and plants?	2
1	41	Name four good sources of plant proteins.	2
1	42	Which are essential amino acids?	2
1	43	What is meant by deamidation of protein?	2
1	44	Write any four significance of Millard reaction.	2
1	45	Write Types of proteolytic enzymes	2
1	46	Write any four application of proteolytic enzymes in foods.	2
1	47	What is meant by Whey protein?	2
1	48	“Proteins are amphiphilic”. Give reasons and write its use.	2
1	49	How egg protein is different from meat protein.	2
1	50	Differentiate between essential and non essential amino acids.	2
1	51	Write the zwitter ion of Lysine and serine amino acids	2
1	52	Give an example of a fruit containing high level of fat.	2
1	53	What are the hydrolysis products of fats and oils?	2
1	54	What is the difference between saturated and unsaturated fatty acids? Give two examples for each.	2
1	55	Define Reichert-Meissl number and iodine number of fats and oils.	2
1	56	How lipids have classified based on its complexity.	2
1	57	How lipids have classified based on its polarity.	2
1	58	How lipids have classified based on its saponification.	2
1	59	What is meant by lipolysis? Explain	2
1	60	Write any two importance of lipid peroxidation process.	2
1	61	Differentiate between lipid peroxidation and thermal decomposition of lipids.	2

1	62	What is meant by Saponifcation and iodine number?	2
1	63	What is meant by frying?	2
1	64	Mention the some changes of food during frying?	2
1	65	Explain Rendered Fat.	2
1	66	Explain the method to Deodorized Oils?	2
1	67	What is meant by hydrogenation of oils?	2
1	68	Give any two differences between hydrogenated oil and hydrogenation of fats.	2
1	69	Write any two advantages inter esterification.	2
1	70	Mention any two general guidelines safety use of oils and fats.	2
1	71	What is Rendered Fat? Give any two examples.	2
1	72	Define Rendered Fat and Rendered Fat.	2
1	73	Write any two best method for lipid analysis?	2
1	74	What is meant by rancidity? Explain	2
1	75	Differentiate between Oxidative and hydrolytic Rancidity.	2
1	76	What is rancidity? Write its types.	2
1	77	What is meant by hydrolytic Rancidity? Explain	2
1	78	What is meant by Oxidative Rancidity? Explain	2
2	79	Differentiate between alcoholic and non- alcoholic beverages.	2
2	80	What is meant by beverages? Explain	2
2	81	Write any four beverages which classified according to their function.	2
2	82	Differentiate between carbonated and non- carbonated beverages.	2
2	83	Differentiate between natural and synthetic beverages.	2
2	84	Differentiate between hot and cold beverages.	2
2	85	Differentiate between stimulating and non-stimulating beverages.	2
2	86	Write any four processes occur during packaged drinking water.	2
2	87	Differentiate between carbonated and mineral water.	2
2	88	What is role of enzymes in fruit juice extraction?	2
2	89	What is meant by still beverages? Explain	2
2	90	Differentiate between energy drink and herbals drink.	2
2	91	What is meant by isotonic drinks?	2
2	92	Write any two differences between hypotonic and hypertonic sport drinks.	2
2	93	Differentiate between low-calorie and dry beverages.	2
2	94	What is meant by broken Tea?	2
2	95	Write any four constituents with composition percentage of Green tea.	2
2	96	Define brick tea and tea dust.	2
2	97	Write any four constituents with content percentage of roasted coffee.	2
2	98	Mention the plantation grades of cured coffee.	2
2	99	Write the composition percentage of protein and caffeine for both tea and coffee.	2
2	100	Write any four constituents with content percentage of roasted coffee.	2

2	101	Write the composition percentage of protein and caffeine for both coffee and cocoa.	2
2	102	What is FSSAI? Write any two functions of FSSAI	2
2	103	What is meant by FSSAI? Explain	2
2	104	Write any four function of FSSAI.	2
2	105	What are the important initiatives taken by FSSAI?	2
2	106	Define juicing and infusion process in manufacturing and packaging processes of beverages.	2
2	107	Define percolation and fermentation process.	2
2	108	Define purification of water and pasteurization process in manufacturing and packaging processes of beverages.	2
2	109	Write any four equipment used in the production of beverages machinery.	2
2	110	Mention the development trends of beverages machinery.	2
2	111	What is meant by reverse osmosis equipment? Explain	2
2	112	Differentiate between ozone sterilizer and ultraviolet sterilizer.	2
2	113	What is meant by isobaric filling machine? Explain.	2
2	114	Differentiate normal and negative pressure filling machine.	2
2	115	Write any four advantages of CIP cleaning system.	2
2	116	Give the difference between sterilization equipment for fluid and canned beverages.	2
2	117	Write a flow chart to represent purification of water.	2
2	118	Define sweeteners and write its classification.	2
2	119	Give any two examples for natural and artificial sweeteners.	2
2	120	Compare the difference between natural and artificial sweeteners.	2
2	121	What is meant by intense sweeteners? Give any two examples.	2
2	122	What is Saccharin? Explain.	2
2	123	What is meant by acidulants? Explain	2
2	124	Define Clarifying agent for beverages	2
2	125	What is clouding agent used in beverages?	2
2	126	Mention any four acidulants used in beverages.	2
2	127	Differentiate between clouding and clarifying agent for beverages.	2
2	128	Write any two examples of natural and synthetic colorants used in beverages industry.	2
2	129	Write any two industrial application of Flavouring agents.	2
2	130	Write a flow chart for preparation of Flavouring emulsions.	2
2	131	Mention any four Flavour used in soft drinks.	2
2	132	How Flavouring agents has been classified? Explain	2
2	133	What is meant by clarity Test?	2
2	134	What is meant by refractive index Test?	2
2	135	Define carbonation.	2
2	136	Mention any two effectiveness of coconut water.	2

2	137	Briefly, explain effectiveness coconut water	2
2	138	Write any two side effect coconut water?	2
2	139	What is meant by sweet toddy? Explain	2
2	140	What are the health benefits of sweet toddy?	2
2	141	Write any four side effect of sweet toddy.	2
2	142	Mention any two allergies and side-effects of sugarcane juice.	2
2	143	What are the uses of sugarcane juice?	2
2	144	Write any four Health Benefits of Sugarcane Juice.	2
2	145	Write any four side effect of coconut milk.	2
2	146	What is meant by coconut milk? Explain.	2
2	147	Write any four nutrition present in coconut milk.	2
2	148	What happens if you drink flavored syrup every day?	2
2	149	What is flavored syrup? Are flavored syrups in your coffee is bad?	2
2	150	Define flavored syrup. What happens if you drink flavored syrup every day?	2
1	151	What is meant by water activity? Explain role and controlling of A_w in food.	5
1	152	Explain the role of water in food.	5
1	153	Summarize a report on Forms of water in food.	5
1	154	Write desirable and undesirable changes take place due to water when food is cooked.	5
1	155	Write a note on any two methods used for removal of bound water from food.	5
1	156	Explain the how water activity is related to sorption Isothermal	5
1	157	Summarize a report on molecular mobility of food	5
1	158	Write a chart on food security. Also, explain food stability.	5
1	159	Explain how the knowledge of food composition helps in formulating a balanced food.	5
1	160	Explain nutrition labelling.	5
1	161	Explain the relationship between water activity and food spoilage.	5
1	162	Write a note on a) Vacuum drying method b) Distillation method.	5
1	163	Discuses a report on sorption Isothermal.	5
1	164	Write a note on a) Sorption Isothermal of food b) Molecular mobility of food.	5
1	165	Explain the food stability and food security.	5
1	166	Explain the effect of cooking on different forms of food.	5
1	167	Explain the changes arise due to effect of heat on the nutrient like carbohydrates and protein present in food?	5
1	168	Explain the changes arise due to effect of heat on the nutrient like protein and fat present in food?	5
1	169	Explain the changes arise due to effect of heat on the nutrient like fat, vitamins present in food?	5

1	170	Explain the changes arise due to effect of heat on the nutrient like protein and vitamins present in food?	5
1	171	Summarize a report on Emulsion form of food	5
1	172	Explain any two chemical reactions of carbohydrates	5
1	173	Write a note on Fischer projection and Howarth cyclic structure of Glucose	5
1	174	Explain the fermentation process or enzymatic reaction of carbohydrates.	5
1	175	Describe the changes occur in carbohydrates, proteins, vitamins, fats and minerals during cooking.	5
1	176	Explain the process of caramelization.	5
1	177	Write a note on Foam or Gels form of food.	5
1	178	Explain hypotonic, hypertonic and isotonic solution. Also, write any two application of osmosis in food processing techniques.	5
1	179	Explain the food dispersion groups with an example.	5
1	180	Describe a report on Emulsion form of food	5
1	181	Write a note on Sols or Gels form of food.	5
1	182	Write a note on foam and explain foam stability and foam destruction.	5
1	183	What are carbohydrates? Explain the properties of monosaccharide	5
1	184	Explain the enzymatic and chemical reactions of carbohydrates.	5
1	185	Discuses a report on caramelization process.	5
1	186	Write the structure of any ten amino acids	5
1	187	Explain protein denaturation.	5
1	188	Explain reaction and properties of protein?	5
1	189	Write a note on salting-in and Salting out.	5
1	190	Describe a report on denaturation and hydrolysis of protein.	5
1	191	Write a note on deamidation, desulfuration and racemization of protein?	5
1	192	Explain Millard Browning and write its significance.	5
1	193	Explain the Functional Roles of Proteins in Foods.	5
1	194	Describe a report on role of protein in food.	5
1	195	Discuse the significance of Maillard reaction in food processing.	5
1	196	Write the six groups of classification of amino acids found in nature. Give an example of each group with its structure.	5
1	197	Explain the classification of protein.	5
1	198	Write a note on a) Protein structure b) Protein Denaturation	5
1	199	Explain proteins of animal, marine and plant origin	5
1	200	Write a note on classification of lipids?	5
1	201	Explain traditional classification of edible fats/oils.	5
1	202	What are lipids? How it's classified? Also, write composition of in cow and buffalo milk.	5
1	203	What are lipids? How it's classified? Write composition of in cow and	5

		human milk.	
1	204	What is meant by lipolysis? Explain lipid peroxidation process.	5
1	205	Describe a report on lipid peroxidation process.	5
1	206	Explain lipid peroxidation process and write its significance.	5
1	207	What is meant by lipolysis? Explain auto-oxidation process of lipids.	5
1	208	Write a note on thermal decomposition of lipids.	5
1	209	Discuses a report on thermal decomposition of lipids and write its health risk.	5
1	210	Write a note on production and processing methods of fats & oil.	5
1	211	Describe a report on processing methods of fats & oil.	5
1	212	Explain production and processing methods of fats & oil.	5
1	213	Briefly explain the modification oil/fat.	5
1	214	Explain the process of hydrogenation of oils.	5
1	215	Discuses a report on inter esterification.	5
1	216	Explain the general guidelines safety use of oils and fats.	5
1	217	Write a note on oil deep-frying safety.	5
1	218	What is Shortenings? Explain the safety measure taken in oil deep-frying.	5
1	219	Explain inter esterification of oils/fats and write its advantages.	5
1	220	Describe a note on chemical reactions of fats.	5
1	221	Write a note on Rancidity and its types.	5
1	222	What is rancidity? Explain the difference between Oxidative and hydrolytic Rancidity.	5
1	223	Explain the following chemical reaction of oils a)Acetylation b)Winterization.	5
1	224	Write a note on Deterioration of Fats.	5
1	225	Describe a report on oxidative rancidity	5
2	226	Explain the classification of beverages with an example.	5
2	227	Describe a report on classification of beverages.	5
2	228	Summarize a report on health importance of beverages.	5
2	229	Write a note on i) Beer ii) Spirit	5
2	230	Describe a brief report on a) Stimulating and non-stimulating beverages b) Carbonated and non-carbonated beverages.	5
2	231	Summarize a report on alcoholic beverages.	5
2	232	What is carbonated water? Write note on packaged drinking water.	5
2	233	What is mineral water? Explain the process of packaged drinking water.	5
2	234	Explain the significance of any five important steps in fruit juice production.	5
2	235	Write a brief report on a) Synthetic beverages b) still beverages	5
2	236	Summarize a report on sports drinks and its classification.	5
2	237	Write a note on a) Alcoholic beverages b) Fruit beverages.	5

2	238	Explain Sport drink and dry beverages.	5
2	239	Write a note on composition and graduation with content percentage of tea.	5
2	240	Describe a report on classification of grades of tea.	5
2	241	Explain the composition and graduation with content percentage of green tea.	5
2	242	Write a note on composition with content percentage of roasted coffee.	5
2	243	Describe a report on the composition of roasted coffee.	5
2	244	Explain the composition of tea and coffee.	5
2	245	Write the composition of roasted coffee and cocoa beans.	5
2	246	Describe a report on composition of roasted coffee and cocoa beans.	5
2	247	Describe a report on FSSAI specification.	5
2	248	Write a note on FSSAI	5
2	249	Explain the function of FSSAI.	5
2	250	Summarize a report on initiatives taken by FSSAI?	5
2	251	Write a note on manufacturing and packaging processes of beverages.	5
2	252	Describe a report on manufacturing and packaging processes of beverages.	5
2	253	Describe a report on packaging processes of beverages.	5
2	254	Explain water filtration equipment used in beverages industry.	5
2	255	Discuss a report on sterilization equipment used in beverages machinery.	5
2	256	Write note on electro dialysis and reverse osmosis equipment.	5
2	257	Explain the physical and chemical sterilization of beverage sanitation.	5
2	258	Describe a report on automatic bottle washing machine used in beverages industry.	5
2	259	Explain the bottle filling machine used in beverages industry.	5
2	260	Discuses CIP Cleaning system and write advantage and disadvantage of this system.	5
2	261	Write a note on CIP Cleaning system used in beverages industry.	5
2	262	Write a note on purification of water with its flow chart.	5
2	263	Describe a brief report on Sweeteners and its classification	5
2	264	What are sweeteners? Write its classification. Mention the relative sweetness of any four natural and artificial sweeteners.	5
2	265	Briefly, explain sweeteners and its classification with examples.	5
2	266	Write a note on synthetic sweeteners by taking any two examples.	5
2	267	Write a note on natural sweeteners by taking any two examples.	5
2	268	Write a note on acidulants used in beverages.	5
2	269	Explain clouding and clarifying agent for beverages	5
2	270	Describe a report on Clouding agent used in beverages	5
2	271	Summarize a brief report on colorants used in beverages industry.	5
2	272	Explain the flavouring agents used in beverages industries.	5

2	273	Write note on flavouring agents and its classification	5
2	274	Explain the methods used for isolation of Flavours.	5
2	275	Explain the quality control tests for syrups.	5
2	276	Write a note on quality control tests a) Clarity Test b) Refractive Index Test c) Identification Test d) Viscosity Determination e) p^H Test	5
2	277	Explain uses, effectiveness and side effect of coconut water?	5
2	278	Write a note on miscellaneous beverages like coconut water.	5
2	279	Describe a report on coconut water.	5
2	280	Define sweet toddy and explain its preparation.	5
2	281	Summarize a report on nutrition facts and potential side effects of palm wine.	5
2	282	Explain the health benefits, Nutrition facts and side effects of sweet toddy.	5
2	283	Explain the nutritional fact, health benefits and side-effects of sugarcane juice.	5
2	284	Write a note on Sugar cane juice.	5
2	285	Describe a report on health benefits, allergies and side-effects of sugarcane juice.	5
2	286	Write a note on coconut milk.	5
2	287	Summarize the report on nutrition facts and side effects of coconut milk.	5
2	288	Discusses a brief report on coconut milk.	5
2	289	Describe a report on flavored syrup.	5
2	290	Briefly explain the syrup and its classification.	5
2	291	Write a note on flavored syrup.	5

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3 **Question Paper Pattern- Model Question Paper** **Q.P Code: 85343**

ST. PHILOMENA'S COLLEGE (AUTONOMOUS), MYSORE			
III SEMESTER –COURSE M.SC FINAL EXAMINATION			
SUBJECT : CHEMISTRY			
TITLE: CHEMISTRY OF FOOD AND BEVERAGES (SC)			
TIME: 3 HRS			**MAX MARKS: 70**
PART-A			
1		**Answer any ten of the following:**	**10x2=20**
	a	Name three scientists who have done pioneering work in the development of	2
	b	Mention any two role of A_w in food processing.	2
	c	Li What are the advantages of vaccume drying over conventional oven drying	2
	d	Write the zwitter ion of Lysine and serine amino acids	2
	e	What is FSSAI? Write any two functions of FSSAI	2
	f	Differentiate between Oxidative and hydrolytic Rancidity.	2
	g	What is meant by deamidation of protein?	2
	h	Explain the method to Deodorized Oils?	2
	i	Write any four application of proteolytic enzymes in foods.	2
	j	Write a flow chart for preparation of Flavouring emulsions.	2
	k	Define flavored syrup. What happens if you drink flavored syrup every day?	2
	l	Write any four side effect of sweet toddy.	2
		Answer any five of the following:	5 x 10= 50
2	a	Explain the how water activity is related to sorption Isothermal	5
	b	Explain the changes arise due to effect of heat on the nutrient like carbohydrates and protein present in food?	5
3	a	Write a note on Fischer projection and Howarth cyclic structure of Glucose	5
	b	Write the six groups of classification of amino acids found in nature. Give an example of each group with its structure.	5
4	a	Explain lipid peroxidation process and write its significance.	5
	b	Explain inter esterification of oils/fats and write its advantages.	5
5	a	Summarize a report on initiatives taken by FSSAI?	5

	b	Write a note on CIP Cleaning system used in beverages industry.	5
6	a	Discuses a report on caramelization process.	5
	b	Write a note on deamidation, desulfuration and racemization of protein?	5
7	a	Describe a report on manufacturing and packaging processes of beverages.	5
	b	Explain clouding and clarifying agent for beverages	5
8	a	Write a note on quality control tests a) Clarity Test b) Refractive Index Test c) Identification Test d) Viscosity Determination e) p^H Test	5
	b	Summarize the report on nutrition facts and side effects of coconut milk.	5

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6 **Question Paper Pattern- Model Question Paper** **Q.P Code: 85343**

ST. PHILOMENA'S COLLEGE (AUTONOMOUS), MYSORE			
III SEMESTER –COURSE M.SC FINAL EXAMINATION			
SUBJECT : CHEMISTRY			
TITLE: CHEMISTRY OF FOOD AND BEVERAGES (SC)			
TIME: 3 HRS			**MAX MARKS: 70**
PART-A			
1		**Answer any ten of the following:**	**10x2=20**
	a	Name three scientists who have done pioneering work in the development of	2
	b	Mention any two role of A_w in food processing.	2
	c	Li What are the advantages of vaccume drying over conventional oven drying	2
	d	Write the zwitter ion of Lysine and serine amino acids	2
	e	What is FSSAI? Write any two functions of FSSAI	2
	f	Differentiate between Oxidative and hydrolytic Rancidity.	2
	g	What is meant by deamidation of protein?	2
	h	Explain the method to Deodorized Oils?	2
	i	Write any four application of proteolytic enzymes in foods.	2
	j	Write a flow chart for preparation of Flavouring emulsions.	2
	k	Define flavored syrup. What happens if you drink flavored syrup every day?	2
	l	Write any four side effect of sweet toddy.	2
		Answer any five of the following:	5 x 10= 50
2	a	Explain the how water activity is related to sorption Isothermal	5
	b	Explain the changes arise due to effect of heat on the nutrient like carbohydrates and protein present in food?	5
3	a	Write a note on Fischer projection and Howarth cyclic structure of Glucose	5
	b	Write the six groups of classification of amino acids found in nature. Give an example of each group with its structure.	5
4	a	Explain lipid peroxidation process and write its significance.	5
	b	Explain inter esterification of oils/fats and write its advantages.	5
5	a	Summarize a report on initiatives taken by FSSAI?	5

	b	Write a note on CIP Cleaning system used in beverages industry.	5
6	a	Discuses a report on caramelization process.	5
	b	Write a note on deamidation, desulfuration and racemization of protein?	5
7	a	Describe a report on manufacturing and packaging processes of beverages.	5
	b	Explain clouding and clarifying agent for beverages	5
8	a	Write a note on quality control tests a) Clarity Test b) Refractive Index Test c) Identification Test d) Viscosity Determination e) p^H Test	5
	b	Summarize the report on nutrition facts and side effects of coconut milk.	5

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